

Alex Huang

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EDUCATION

Yale University | 2021-2025 | Combined B.S./M.S. in Computer Science | GPA: 3.9

Classes: Computer Architecture, Neural Nets(EE Hardware), Data Structures, Linear Algebra, Systems Programing, Embedded Systems, Algorithms, Circuits, Intro Computer Engineering, Compilers, Networked Systems, OS

EXPERIENCE

Software Developer Intern | [Neurobit](#) | Taipei, Taiwan | Part-time with Study-Abroad May 2023 - August 2023

- Improved gyroscope to video calibration to diagnose eye diseases at a [Taiwanese Biomedical Tech startup](#).
- Designed Blazor software to label eye tracking data to improve the gaze Pytorch model, labeling 3000 samples.

Undergraduate Researcher | [Interactive Machines Group](#) @ Yale University | New Haven, CT May 2022 - May 2023

- Led a project in the summer to develop new behaviors for our robot [Shutter](#), using ROS and MoveIT.
- Modified codebase to allow two Kinect sensors by establishing **identification coherence** using **PyTorch** and heuristics, increasing the effective range of Shutter by 85%
- Deployed Shutter in “in-the-wild studies” to research effective ways of engaging interaction, resulting in a paper.
- Created a real-time interface to evaluate robot status and performance with Vue.js and websockets.

Full-Stack Developer Intern | [Doordash](#) | New York City, NY May 2021 - August 2021

- Prototyped Google Maps Integration for chain vendors and new menu search features using React and Django Backend.
- Migrated and refactored Angular codebase into React while resolving critical caching and XSS bugs preventing launch.
- Developed E2E front-end testing protocols/documentation using Cypress.
- Implemented advanced analytics tracking using Amplitude to track online ordering preferences.

Lead Full-Stack Developer and Architect | [Gapyearly](#) | New York City, NY January 2021 - May 2021

- Self-Architected and maintained [custom web app project](#) from scratch with a UX designer for a student run organization.
- Built features including real-time chat, user matching, profiles, and a blog using Next.JS/React for over 100+ users..
- Developed a Back-End Stack using Strapi and MongoDB. Stored images in an S3 bucket. Implemented SEO

LEADERSHIP & STUDENT COMMUNITY

Yale Radio (WYBCx) | [Head of Broadcast Engineering Department](#) January 2021 - Present

- Self-built a new studio for our show, “The Moon”, bringing in bands and recording live studio shows, along with live mixing all shows.

Yale I-EEE | [Treasurer](#) September 2021 - Present

- Organized over 10000\$ of funding for events, trips, and equipment for our 7 project teams.
- Formerly worked on the 6502 team, building a computer from breadboards and 6502 microchip.

PROJECTS

github.com/NukeWolf

CPSC 520 Parcv2 Processor / Designed a 5-stage pipelined bypassing processor in Verilog for Parcv2, a variant of MIPS.

Implemented 2 instruction fetching and out-of-order execution with a reorder buffer

WeenyOS Projects / Implemented basic Page Allocation for process and a heap management system using optimistic memory allocation using C.

HTTP1.1 Select Server / Created a HTTP server in Java from scratch using a selector server model with epoll. Features include a multithreaded worker pool, CGI execution, header parsing, chunked transfer

Rucio @ CERN / Contributed open-source code to improve data transfer times, through parallel data streams

Arduino LightSaber / Physical Lightsaber built with Arduino and Leds, using a gyroscope to detect motion.

SKILLS

Coding Languages | Python, JS, HTML, CSS, C, C++, SQL, Verilog, Java, C#,

Tools | Vivado, React, ROS, Blazor, Git, UNIX, C#, Linux, Angular, MoveIt, Xilinx

Skills | Micro-Architecture, Cyber/CTF, Agile Development, Digital Systems, Robotics